Indian Institute of Technology Dharwad
Department of EECE
EE 601L: Neural Networks and Deep Learning Lab

LAB-6 **24/02/ 2026**

Objective

1. To understand sequence modeling using Recurrent Neural Networks.
2. To implement an RNN from scratch for next token prediction.
3. To represent text tokens numerically using indexing and one-hot encoding.
4. To implement forward pass, backward pass (BPTT), and optimization.
5. To address the exploding gradient problem using gradient clipping.

Task 1: Dataset Generation

1. Generate synthetic sequences.
2. Include token EOS at end of each sequence.
3. Display at least one sample sequence.

Task 2: Token Representation

1. Create word_to_idx and idx_to_word dictionaries (include UNK).
2. Display vocabulary size and total number of sequences.

Task 3: Train–Test Split

1. Split dataset into 80% training and 20% testing.
2. Generate input sequences and shifted target sequences.

Task 4: One-Hot Encoding

1. Implement one-hot encoding function.
2. Encode a token and a full sequence.
3. Verify output shape: (sequence_length, vocab_size, 1).

Task 5: RNN Implementation (From Scratch)

Implement:

1. Forward Pass
2. Backward Pass (Backpropagation Through Time)
3. Loss calculation (Cross-Entropy)
4. Parameter update using Gradient Descent
5. Gradient Clipping to handle exploding gradients

Task 6: Training

1. Write complete training loop.
2. Train for multiple epochs.
3. Print training loss per epoch.
4. Evaluate on test data.

Expected Observations

1. RNN learns sequence dependency between a and b.
2. Loss decreases across epochs.
3. Without gradient clipping → exploding gradients may occur.
4. With clipping → stable training.

Inference and Conclusion

Students must clearly state:

1. Learning behavior of RNN
2. Effect of gradient clipping
3. Importance of sequential memory
4. Limitations of vanilla RNN

Problem 2: RNN for Time Series Prediction (Sine Wave)

Objective

To use RNN for time-series forecasting.

Problem Description

Given:

- A sine wave sequence of 50 consecutive values

Task:

Predict the 51st value of the sine wave.

Tasks

1. Generate sine wave dataset.
2. Create input sequences of length 50.
3. Implement RNN for regression.
4. Train the model.
5. Predict the next value.
6. Compare predicted value with actual sine value.
7. Plot actual vs predicted output.

Expected Outcome

1. RNN should capture periodic behavior.
2. Prediction error should reduce with training.
3. Demonstrates RNN capability for time-series regression.

Submission Instructions (Strict)

1. Implement each task separately.
2. Do NOT combine tasks into one code block.
3. Follow the exact task order.
4. Write:
    - Objective
    - Hypothesis
    - Experimental Description
    - Observations
    - Conclusion